J. Dairy Sci. 109:6364–6375
https://doi.org/10.3168/jds.2026-28267

A systematic mapping of empirical studies of milking and operator efficiency in dairy farming

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ABSTRACT

The current challenges associated with increasing global demand for dairy products, developing herd structures, and declining availability of labor have resulted in the need for improvements in contemporary dairy farm infrastructure and management practices. Internationally, across different production systems, the milking process accounts for between 33% and 75% of annual labor requirements on commercial dairy farms. Because milking efficiency is multifactorial and highly context-specific, a clearer understanding of the factors that influence it—as evaluated through key performance indicators—is required to support future optimization strategies. This study conducted a systematic review of the milking efficiency literature to identify knowledge gaps and evaluate how previous research approached this topic. Five research questions (RQ) guided the review, relating to contributions by country and production system (RQ1), problem types and research objectives (RQ2), methodologies used (RQ3), milking efficiency values across parlor systems (RQ4), and the effect of operator involvement (RQ5). Fifteen publications were analyzed, representing 23 sample groups and 315 parlors. Outcome variables included cows milked per hour (cows/h), cows milked per operator per hour (cows/h per operator), cows milked per cluster per hour (cows/h per cluster), and clusters per operator (clusters/operator). Four parlor types were represented: herringbone double-up, parallel double-up (P-DU), herringbone swing-over (H-SO), and rotary (RO). The United States and confinement-based systems contributed the most studies. Comparative research was the dominant problem type, and time-and-motion analysis was the most frequently used methodology. The P-DU parlors achieved the greatest overall throughput (352 cows/h), RO parlors achieved the highest operator-level efficiency (143 cows/h per operator) and clusters handled per operator (27 clusters/operator), and H-SO parlors achieved the highest throughput per cluster (5.6 cows/h per cluster). Across all parlor types, increasing operator number did not significantly increase hourly cow throughput or cows/h per cluster, but significantly reduced cows/h per operator and clusters/operator at milking. This systematic review synthesizes the structure, methods, and empirical findings of the milking efficiency literature. It provides a summary understanding of milking efficiency levels across a range of parlor types operating in different production systems and quantifies the effect of operator on milking efficiency levels relative to parlor type. For researchers in this field, the authors of this review have highlighted key directions for future investigation into milking efficiency optimization through the alignment of parlor infrastructure, automation specification, and management practices at milking.
Key words: milking efficiency, rotary, herringbone swing-over, herringbone double-up, parallel double-up

INTRODUCTION

Rising global demand for dairy products, evolving herd structures, and persistent shortages in labor have converged to emphasize the need for innovation in dairy farm infrastructure and management (Barkema et al., 2015, Nye, 2018, OECD/FAO, 2023). Globally, commercial milk production is broadly classified into 2 production systems: (1) confinement and (2) pasture-based systems (Powell et al., 2013). Confinement systems, predominant in major milk producing countries such as the United States, India, and Germany, house cattle indoors year-round and typically provide TMR that support higher milk yields (Arnott et al., 2015). In contrast, pasture-based systems allow cattle to graze outdoors on fresh grass for most of their diet and time. These systems are common in countries where ideal climate conditions allow for effective grass utilization (e.g., Ireland, New Zealand; Ramsbottom et al., 2015).

Received January 9, 2026.
Accepted February 24, 2026.
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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Across both systems, the milking process accounted for between 33% and 75% of annual labor requirements on commercial dairy farms (Auernhammer, 1989, O'Donovan et al., 2008, Deming et al., 2018). Within the EU, approximately 2.5 million workers left the Agri-industry between 2007 and 2017 (Ryan, 2023). Consequently, milking efficiency has emerged as a central focus of global dairy science, as farm scale and technological complexity continue to increase in both pasture- and confinement-based systems.

Most milking systems used internationally are conventional parlors, including linear designs (herringbone and parallel) and rotary platforms. The farm-to-farm variability in the number of cows milked per hour is primarily driven by differences in herd milking characteristics, parlor type, parlor size, automation specification, and management practices at milking (Prendergast et al., 2024b). Research indicates that rotary systems deliver higher levels of milking efficiency compared with herringbone parlors (Edwards et al., 2013a,b). However, increases to parlor size will have a greater effect on enhancing cow throughput for herringbone parlors rather than rotaries (Prendergast et al., 2024a). Furthermore, a second operator at milking was found to have no effect on rotary parlor milking efficiency, but their merits may be found in a herringbone milking scenario (Prendergast et al., 2024a). Milking efficiency is multifactorial and context-specific. Understanding its drivers, particularly those linked to parlor technology and operator management, will be critical to sustaining the vitality of dairy farming worldwide.

The future direction of global milking efficiency research can be guided by a systematic review of prior study motivations, hypotheses, methodologies, and evaluation metrics. Despite extensive research, a fragmented understanding remains with respect to milking efficiency across different production systems, countries, and operational scales. To the authors' knowledge, no comprehensive synthesis exists that compares assessment methods, operational characteristics, and efficiency outcomes across milking parlor types. Consequently, key knowledge gaps persist regarding how efficiency is defined and measured, how parlor designs perform relative to each other, and how operator factors influence results. Therefore, the objective of this systematic review was to address these gaps by synthesizing empirical evidence to clarify where the research direction has concentrated, the methodologies used, and how milking efficiency differs across parlor types depending on the number of operators. Addressing these gaps will guide industry stakeholders toward future research priorities and actionable strategies to optimize milking performance.

Specifically, the authors aimed to answer the research questions (**RQ**) in the following list:

- RQ1: Which countries and production systems contribute most to the milking efficiency literature?
- RQ2: What problem types and research objectives are most commonly addressed?
- RQ3: Which methodologies are most frequently employed to research milking efficiency?
- RQ4: How do milking efficiency values compare among different parlor systems?
- RQ5: How does operator involvement affect milking efficiency across parlor designs?

MATERIALS AND METHODS

Literature Search

A literature search was conducted using 2 online databases: ScienceDirect (https://www.sciencedirect.com/) and Google Scholar (https://scholar.google.com/). The search strategy included the use of specific keywords, primarily "milking parlor efficiency," "herringbone swing-over parlor efficiency," "rotary parlor efficiency," "herringbone double-up parlor efficiency," "parallel parlor milking efficiency," and "milking parlor productivity." Authors that were known to have published relevant literature were also searched directly so their publication list could also be examined for relevance. Reference lists from published papers were also reviewed for any additional studies relevant to the review. The objectives of each reviewed study were categorized to a specific "problem type" based on the criteria detailed in Table 1. A summary of the dataset indicating the research papers'

Table 1. Problem type categories with respective definitions

Problem type	Definition
Descriptive	To define or describe various characteristics and parameters related to various milking parlor types or milking processes
Comparative	To compare or evaluate differences in milking process times or milking efficiency across different milking parlor types or milking management practices
Technical development	To design or demonstrate (or both) methods, systems, or simulations that attempt to optimize milking process efficiency through variations in parlor configurations or management practices at milking

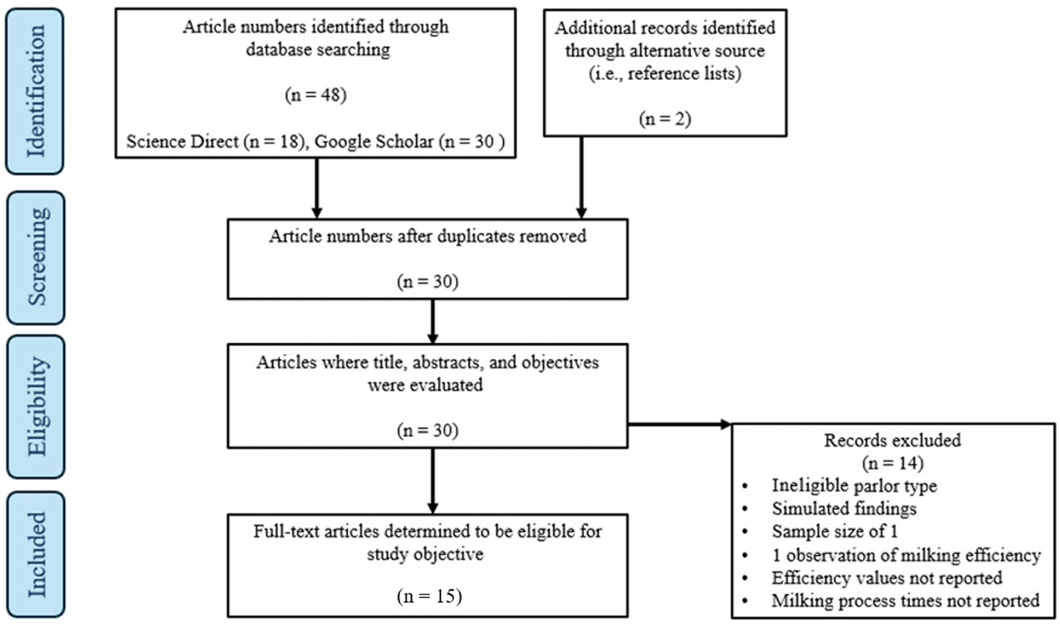

Figure 1. The PRISMA flowchart detailing the literature search and the study collection process.

author(s), year of publication, country of origin, number of farms sampled, and parlor type was compiled (Table 2). The literature search and study selection process were conducted in accordance with the PRISMA 2020 reporting guidelines (Figure 1; Sohrabi et al., 2021). Selected studies ranged between the years of 1973 and 2024.

Eligibility Criteria and Screening

Studies were initially screened by title, abstract, and objective. Research papers that reported milking efficiency values from the direct analysis of empirical data were selected; simulated findings were not selected for review. Only studies that examined herringbone swing-over (**H-SO**), herringbone double-up (**H-DU**), parallel swing-over, parallel double-up (**P-DU**), and rotary (**RO**) milking parlors were selected for review. Studies were further screened for the inclusion of the following criteria: (1) contained original, peer-reviewed research, (2) published as either scientific journal articles or conference papers in the agricultural sciences, (3) reported sample numbers, (4) reported parlor type, and (5) reported milking efficiency (cows/h) values.

Data Extraction

Outcome variables that were specifically examined in this study included (1) cows milked per hour (cows/h), (2) cows milked per operator per hour (cows/h per operator), (3) cows milked per hour per cluster (cows/h per cluster), and (4) clusters per operator at milking (clusters/operator). If outcome variables (2), (3), and (4) were not directly reported in the selected studies, they were calculated using equations [1], [2], and [3]:

$$\text{Cows/h per operator} = \text{(cows/h)/(number of operators at milking)}, \quad [1]$$

$$\text{Cows/h per cluster} = \text{(cows/h)/(number of clusters in the parlor)}, \quad [2]$$

$$\text{Clusters/operator} = \text{(number of clusters in the parlor)/(number of operators at milking)}. \quad [3]$$

The number of rows for H-DU, P-DU, H-SO, and RO parlors were defined as "batch numbers." If batch numbers were not reported directly, they were calculated using equation [4]:

$$\text{Batch numbers} = \text{(herd size at milking)/(number of clusters in the parlor)}. \quad [4]$$

For all relevant outcome variables, milking process time was defined as "first cluster attached until last cluster detached." The raw literature dataset was then screened for outliers, which were defined as being ±3 median absolute deviations from the median and were subsequently removed. The average rate of outlier occurrence per outcome variable was 3% (1%–5%). Upon completion of outlier removal, the dataset was referred to as "refined."

Data and Statistical Analysis

Data regarding herd size, parlor size, number of operators at milking, batches, milkings per day, and milking efficiency metrics were compiled by parlor type and summarized using means and ranges. Milking efficiency metrics were then examined among the different parlor types, with respect to 1- and 2-operator milkings, for quartiles based on cows/h per operator.

Weighted linear mixed-effects models were developed to evaluate the effects of parlor type, production system, cluster number, and operator number on milking efficiency key performance indicators (**KPI**). Parlor type and production system were treated as categorical fixed effects, whereas cluster number and operator number were included as continuous fixed effects. Study was included as a random effect to account for clustering of observations within primary studies. Models were weighted by study sample size to account for differences in study-level contribution.

Continuous predictors were standardized (z-score) to allow effect size comparability. Model coefficients for cluster and operator number therefore represent effects per 1 SD increase. Significance of fixed effects was assessed using *t*-tests on model coefficients. Least squares means were computed for parlor type and compared using Tukey-adjusted pairwise contrasts. Statistical significance was declared at $P < 0.05$.

Data processing and statistical analyses were performed using Microsoft Excel 2016 and RStudio (version 2022.07.2; https://posit.co/download/rstudio-desktop/).

RESULTS

Table 2 details the studies selected for review, year of publication, country of origin, number of farms sampled in their research, their system of production, and parlor type examined.

Fifteen studies were included in the analysis, encompassing 23 sample groups and 315 individual parlors in total. Four different parlor types were identified: (1) H-DU (n = 34), (2) P-DU (n = 28), (3) H-SO (n = 36), and (4) RO (n = 217). Confinement-based (**CB**) and pasture-based (**PB**) production systems consisted of 43% and 57% farms in the total sample, respectively. All studies that were associated with CB production originated in the United States, whereas PB research exhibited greater geographic diversity, including studies from Europe (United Kingdom, Ireland, North Continental Europe), Latin America (Mexico), and Oceania (Australia and New Zealand).

Among CB production systems, RO parlors were the most frequently researched (57%), followed by H-DU

Table 2. Summary of studies (n = 15) involved in systematic review detailing their author(s), year of publication, country of origin, number of farms sampled in their research, their system of production, and parlor type examined[1]

Study no.	Citation	Country	Farms	System	Parlor	Clusters
1	Appleman and Micke, 1973	US	4	CB	H-DU	8
2	Armstrong and Quick, 1986	US	13	CB	H-DU	19
3	Chang et al., 1994	US	1	CB	H-DU	26
3	Chang et al., 1994	US	2	CB	P-DU	27
4	Thomas et al., 1996	US	2	CB	P-DU	75
4	Thomas et al., 1996	US	2	CB	H-DU	36
5	Smith et al., 1998	US	24	CB	P-DU	76
5	Smith et al., 1998	US	11	CB	H-DU	49
5	Smith et al., 1998	US	3	CB	RO	37
6	Smith et al., 1999	US	14	CB	RO	47
7	Armstrong et al., 2001	MEX	37	PB	RO	57
7	Armstrong et al., 2001	AUS	4	PB	RO	58
7	Armstrong et al., 2001	US	61	CB	RO	50
8	Ohnstad et al., 2012	UK	3	PB	H-DU	56
8	Ohnstad et al., 2012	UK	1	PB	H-SO	16
9	Ozolins et al., 2012	LAT	5	PB	RO	26
10	Edwards et al., 2013	NZ	61	PB	RO	55
11	Edwards et al., 2013	IRE	19	PB	H-SO	21
12	Andersson, 2014	NCE	9	PB	RO	57
13	Mangalis et al., 2015	LAT	7	PB	RO	46
14	Mangalis and Priekulis, 2018	LAT	7	PB	RO	43
15	Prendergast et al., 2024	IRE	16	PB	H-SO	20
15	Prendergast et al., 2024	IRE	9	PB	RO	50

[1]US = United States, MEX = Mexico, AUS = Australia, UK = United Kingdom, LAT = Latvia, NZ = New Zealand, IRE = Ireland, NCE = North Continental Europe (Sweden, Denmark, Germany), CB = confinement based, PB = pasture based, H-DU = herringbone double-up, P-DU = parallel double-up, H-SO = herringbone swing-over, RO = rotary.

parlors (23%), and P-DU parlors (20%). Research into the use of H-SO in CB systems was not observed. Similarly, RO parlors were most frequently researched (78%) in PB systems followed by H-SO (20%) and then H-DU (2%). Research into the use of P-DU parlors in PB systems was not observed.

Figure 2 details the flow of studies from problem type to publication source to methodologies used among the literature. The most researched problem type classifications among the literature were comparative (53%), descriptive (27%), and technical development (20%). The *Journal of Dairy Scienc*e (https://www.journalofdairyscience.org/) was the most common publication source among the literature (40%), followed by the Western Dairy Management Conference (https://wdmc.org/), Engineering for Rural Development (https://www.mdpi.com/journal/agriculture/special_issues/engineering_rural_development), and other (13%, respectively), followed by American Society of Agricultural and Biological Engineers (https://asabe.org/), *Animal Production Science* (https://connectsci.au/an), *Animal Research* (https://journalanimalresearch.com/), and *Journal of Dairy Research* (https://www.cambridge.org/core/journals/journal-of-dairy-research), which comprise 7% of publication sources.

Figure 3 details the methodologies and the frequency of their use among the milking efficiency literature. Among methodologies, time and motion was used most commonly by researchers (60%); followed by the use of video recordings (20%); use of milking management data, telephone, and infrastructure surveys (13% in total); and access to milk yield databases (13%) and management surveys (7%).

Table 3 details the farm characteristics, management practices, and milking efficiency values for the different parlor types identified in the literature.

The largest herds were observed among farms with P-DU parlors at a value of 1,455 cows, followed by RO (688), H-DU (634), and H-SO parlors (166), respectively. Similarly, P-DU parlors had the largest average parlor size at a value of 72 clusters, followed by RO (49), H-DU (35), and H-SO parlors (20). On average, P-DU required the most operators at milking (3.8), followed by H-DU (2.1), RO (1.9), and H-SO (1.4). The highest average number of milkings per day were carried out on farms with P-DU parlors (2.8), followed by H-DU (2.3) and farms with H-SO and RO parlors had an equal average number of milkings per day (2.0).

The P-DU parlors achieved the highest average value for cows/h (352), followed by RO (229), H-DU (152), and H-SO parlors (111). Rotary parlors achieved the highest average value for cows/h per operator (143), followed by P-DU (96), H-SO (82), and H-DU (69). The H-SO parlors achieved the highest average value for cows/h per cluster (5.6), followed by RO (3.8), H-DU (3.3), and P-DU (1.6). In regard to the number of clusters per operator at milking, RO parlors achieved the highest average value (27.0), followed by P-DU (20.0), H-DU (16.7), and H-SO (13.1).

Table 4 summarizes milking efficiency performance among parlor types for 1- and 2-operator scenarios.

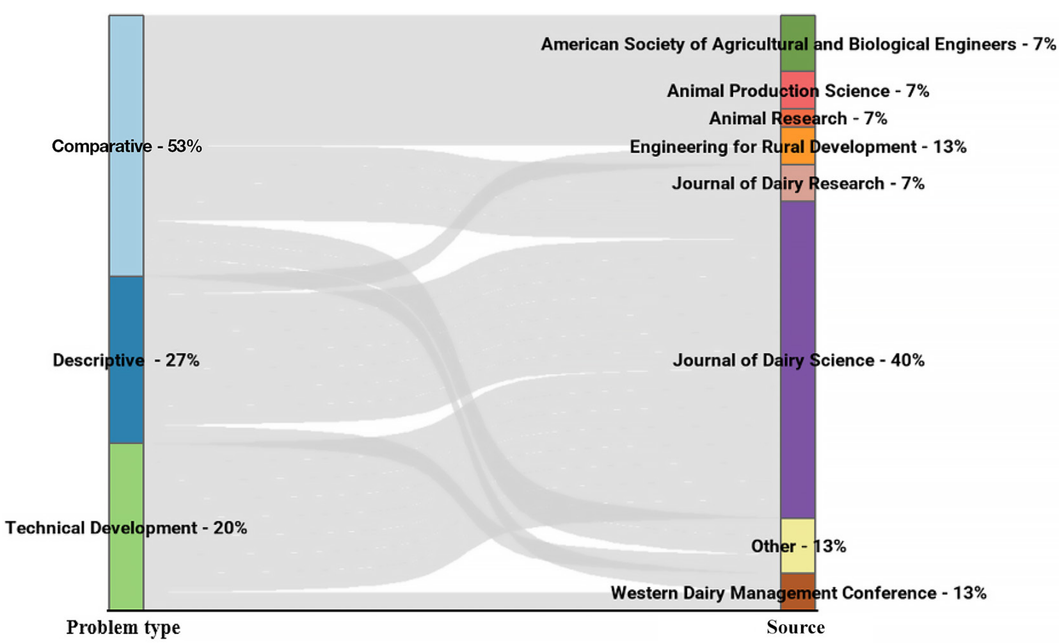

Figure 2. The flow of studies from problem type to publication source to methodologies used among the literature.

Table 3. Farm characteristics, management practices, and milking efficiency values for H-DU, H-SO, P-DU, and RO parlors showing mean (range)

Parameter	Parlor type			
	H-DU	H-SO	P-DU	RO
Herd size (cows)	634 (177–1,712)	166 (45–360)	1,455 (220–2,838)	688 (208–2,000)
Clusters in parlor	35 (4–90)	20 (9–36)	72 (20–100)	49 (20–80)
Operators at milking	2.1 (1.0–7.0)	1.4 (0.9–3.1)	3.8 (1.0–7.0)	1.9 (1.0–5.0)
Batches	18 (4–43)	8 (4–16)	22 (8–36)	13 (6–25)
Milkings per day	2.3 (2.0–4.0)	2.0 (2.0–2.0)	2.8 (2.0–4.0)	2.0 (2.0–4.0)
Cows/h	152 (31–465)	111 (42–226)	352 (99–610)	229 (83–536)
Cows/h per operator	69 (31–116)	82 (34–126)	96 (51–136)	143 (47–434)
Cows/h per cluster	3.3 (0.7–9.7)	5.6 (3.1–8.9)	1.6 (0.7–5.0)	3.8 (1.0–7.2)
Clusters/operator	16.7 (4.0–48.0)	13.1 (5.1–24.2)	20.0 (11.4–28.1)	27.0 (10.3–72.0)

Among 1-operator milkings, RO parlors achieved the highest average throughput (Q4 = 240 cows/h), followed by H-SO (126 cows/h), P-DU (117 cows/h), and H-DU (100 cows/h). When 2 operators were present, the ranking remained similar, with RO parlors again showing the highest mean throughput (Q4 = 536 cows/h). On a per-operator basis, RO parlors maintained the greatest efficiency (up to 434 cows/h per operator), reflecting the scale and automation advantage of these designs.

Table 5 presents the weighted mixed models for milking efficiency KPI by parlor type, production system, cluster number, and operator number at milking. For milking throughput (cows/h), cluster number was the strongest predictor ($\beta = 89.8$, $P < 0.05$). The cluster estimate of +89.8 represents the effect of a 1 SD increase in cluster number. Given an SD of 18.9 clusters, this corresponds to approximately +4.8 cows/h per additional cluster. The P-DU parlors achieved 190.3 fewer cows/h

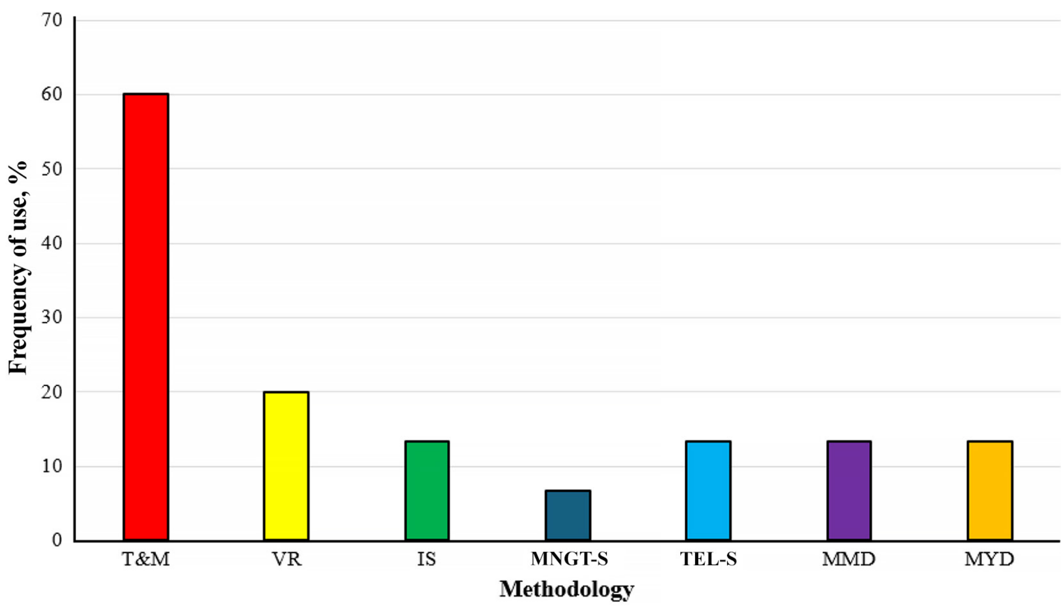

Figure 3. Bar chart detailing methodologies and the frequency of their use among the literature. T&M = time-and-motion analysis, VR = video-recording, IS = infrastructure survey, MNGT-S = management survey, TEL-S = telephone survey, MMD = milking management data, MYD = milk yield database.

Table 4. Comparison of milking efficiencies among different parlor types, between 1- and 2-operator milkings, for quartiles based on cows/h per operator[1]

		1 operator		2 operators	
Parlor	Parameter	Q1	Q4	Q1	Q4
H-SO	Cows/h	98	126	85	226
	Cows/h per operator	90	126	58	113
	Cows/h per cluster	5	9	4	8
	Clusters/operator	16	24	9	18
H-DU	Cows/h	60	100	94	165
	Cows/h per operator	60	100	47	83
	Cows/h per cluster	4	10	2	3
	Clusters/operator	11	48	10	36
P-DU	Cows/h	99	117	274	352
	Cows/h per operator	99	117	91	117
	Cows/h per cluster	4	5	2	3
	Clusters/operator	23	28	20	23
RO	Cows/h	123	240	181	536
	Cows/h per operator	123	240	103	434
	Cows/h per cluster	3	5	3	7
	Clusters/operator	36	72	18	40

[1]Q1 = quartile 1, lower 25% of dataset, Q4 = quartile 4, upper 25% of dataset, H-DU = herringbone double-up, P-DU = parallel double-up, H-SO = herringbone swing-over, RO = rotary.

($P < 0.05$) compared with H-DU parlors. The H-SO and RO parlors did not significantly differ ($P = 0.68$, $P = 0.93$, respectively) from the H-DU intercept reference. The effect of PB system on cows/h was not significant ($P = 0.39$). Operator number did not have a significant effect on cows/h ($P = 0.51$).

Clusters significantly increased cows/h per operator values ($P < 0.005$), with the estimate of +52.3 being equivalent to +2.8 cows/h per operator per additional cluster. In contrast, each additional operator significantly decreased cows/h per operator values ($P < 0.005$), with the estimate of −41.7 representing the effect of 1 SD decrease in operator number. Given an SD of 0.94 operators, this corresponds to an effect of −44 cows/h per operator per additional operator at milking. Rotary parlors exhibited near-significant advantage in cows/h per operator values over H-DU ($\beta = 36.7$, $P = 0.06$). Herringbone swing-over, P-DU, and the effect of PB system on cows/h per operator was not significant (all $P > 0.05$). For cows/h per cluster, no significance level was reached by any of the examined variables (all $P > 0.05$).

Clusters significantly increased values for clusters/operator ($P < 0.0005$), with an estimate of +11.2 being equivalent to +0.59 clusters/operator per additional cluster. In contrast, operator number significantly decreased the number of clusters/operator at milking ($P < 0.0005$), with an estimate of −11.5 being equivalent to −12.1 clusters/operator per additional operator added at milking.

Table 5. Weighted mixed models for milking efficiency KPI by parlor type, production system, cluster number, and operator number at milking[1]

KPI	Parameter	Estimate	t ratio[2]	P-value
Cows/h	(Intercept)	224.4	6.4	<0.005
	H-SO	27.6	0.4	0.68
	P-DU	−190.3	−2.9	<0.05
	RO	4.4	0.1	0.93
	System-Past	−43.8	−0.9	0.39
	Clusters	89.8	3.0	<0.05
	Operators	18.4	0.7	0.51
Cows/h per operator	(Intercept)	89.2	5.5	<0.0005
	H-SO	43.5	1.5	0.17
	P-DU	−5.0	−0.2	0.81
	RO	36.7	2.1	0.06
	System-Past	−22.1	−0.9	0.39
	Clusters	52.3	3.5	<0.005
	Operators	−41.7	−3.0	<0.005
Cows/h per cluster	(Intercept)	3.5	6.4	<0.005
	H-SO	1.5	1.7	0.12
	P-DU	−0.4	−0.8	0.47
	RO	0.1	0.2	0.88
	System-Past	−0.6	−0.7	0.47
	Clusters	−0.3	−0.7	0.52
	Operators	−0.2	−0.5	0.64
Clusters/Operator	(Intercept)	22.2	8.7	<0.0005
	H-SO	−0.7	−0.2	0.84
	P-DU	−0.6	−0.4	0.69
	RO	−1.2	−0.9	0.38
	System-Past	1.3	0.3	0.74
	Clusters	11.2	5.5	<0.0005
	Operators	−11.5	−5.3	<0.0005

[1]P-DU = parallel double-up, H-SO = herringbone swing-over, RO = rotary, System-Past = PB system.

[2]The ratio of the fixed-effect estimate to its respective standard error.

Table 6. Least squares means comparison of milking efficiency KPI by parlor type[1]

KPI	Comparison		Estimate	t ratio[2]	P-value
	Parlor 1	Parlor 2			
Cows/h	H-DU	H-SO	−27.6	−0.4	0.977
	H-DU	P-DU	190.3	2.7	<0.05
	H-DU	RO	−4.4	−0.1	1.000
	H-SO	P-DU	217.9	2.3	0.095
	H-SO	RO	23.2	0.4	0.983
	P-DU	RO	−194.7	−3.5	<0.05
Cows/h per operator	H-DU	H-SO	−43.5	−1.4	0.514
	H-DU	P-DU	5.0	0.2	0.996
	H-DU	RO	−36.7	−2.0	0.198
	H-SO	P-DU	48.5	1.1	0.684
	H-SO	RO	6.8	0.2	0.995
	P-DU	RO	−41.7	−1.7	0.343
Cows/h per cluster	H-DU	H-SO	−1.5	−1.5	0.426
	H-DU	P-DU	0.4	0.7	0.903
	H-DU	RO	−0.1	−0.1	0.999
	H-SO	P-DU	1.9	1.3	0.542
	H-SO	RO	1.4	1.6	0.391
	P-DU	RO	−0.4	−0.6	0.937
Clusters/operator	H-DU	H-SO	0.7	0.2	0.997
	H-DU	P-DU	0.6	0.4	0.980
	H-DU	RO	1.2	0.9	0.825
	H-SO	P-DU	0.0	0.0	1.000
	H-SO	RO	0.5	0.1	0.999
	P-DU	RO	0.5	0.2	0.996

[1]H-DU = herringbone double-up, P-DU = parallel double-up, H–SO = herringbone swing-over, RO = rotary.

[2]The ratio of the fixed-effect estimate to its respective standard error.

Significances from the effect of parlor type and system of production on clusters/operator were not detected (all $P > 0.05$).

Table 6 shows a LSM comparison of milking efficiency KPI by parlor type. Comparisons confirmed that P-DU achieved significantly lower cow throughput values than H-DU parlors (−190 cows/h, $P < 0.05$) and RO parlors (−195 cows/h, $P < 0.05$). No other significant parlor type differences were detected across any measured KPI (all $P > 0.05$).

DISCUSSION

Overview of Key Findings

The review and analysis of selected literature confirmed that P-DU parlors achieved the highest values for cows milked per hour (Table 3) with an average of 352 cows/h, which was 54%, 132%, and 217% greater than values for RO, H-DU, and H-SO parlors, respectively. This difference in throughput is likely driven by the larger parlor size documented among P-DU parlors, with an average of 72 clusters, which was 47%, 106%, and 260% greater than values for RO, H-DU, and H-SO parlor sizes, respectively. Additionally, P-DU parlors had a greater number of operators present at milking at an average of 3.8, which was 81%, 100%, and 171% greater than values for H-DU, RO, and H-SO, respectively. Hence, although P-DU parlors were found to be the most efficient parlor among the literature in terms of cows milked per hour, the high throughput value of this system was likely facilitated by the large parlor size coupled with the large number of operators.

With respect to operator efficiency, RO parlors achieved the greatest value for cows milked per operator per hour, at an average of 143 cows/h per operator, which was 49%, 74%, and 107% greater than values observed for P-DU, H-SO, and H-DU parlors, respectively. In addition, RO parlors also achieved the highest value for the number of clusters per operator at milking, at an average of 27, which was 35%, 62%, and 106% greater than the averages observed for P-DU, H-DU, and H-SO parlors, respectively. Hence, RO parlors were found to be the most efficient in terms of cows milked per operator per hour. The RO parlors also achieved the highest number of clusters per operator. Together, these findings highlight the RO parlors' ability to optimize labor use during milking. Among the literature, H-SO parlors achieved the highest value for cows/h per cluster, with an average of 5.6, which was 47%, 70%, and 250% greater than averages observed for RO, H-DU, and P-DU parlors, respectively (Table 3). Therefore, despite having lower overall cow throughput and cows milked per operator per hour values in comparison to other parlor types, H-SO parlors achieved the highest milking efficiency per cluster. This highlights their capacity for infrastructure optimization and greater cluster utilization across parlor types.

Effect of Operator Number and Management Practices

We found that the presence of an additional operator at milking did not significantly (P = 0.51) affect cow throughput per hour across parlor type (Table 5). This aligns with a previous observation by Prendergast et al. (2024b), who found that a second operator at milking did not significantly affect milking process time efficiency for H-SO and RO parlors. Increasing operator number did not significantly increase overall cow throughput per hour, but significantly reduced cows/h per operator and clusters/operator values (Table 5). These findings suggest that increasing operator number does not necessarily improve parlor efficiency within the defined milking process time window. This theory is built upon data that observed milking process time being defined as "first cluster attached until last cluster detached." Hence, the value of a second operator at milking could be greater when used outside of this time frame (e.g., setup time, cleanup time; Prendergast et al., 2024b).

Findings generated from this systematic review reinforce those reported in individual studies. For example, the authors of this paper observed that RO parlors achieve, on average, higher levels of milking efficiency than H-SO parlors (Edwards et al., 2013a,b; Edwards et al., 2020; Prendergast et al. 2024a,b). The authors of this paper also found that H-SO parlors achieved the greatest average values for cows/h per cluster when compared with other parlor types. This observation aligns with previous research that determined a greater effect estimate from cluster number on H-SO milking efficiency when compared with other parlor types (Prendergast et al. (2024a,b).

Considerable variances in efficiencies were observed between quartiles 1 (**Q1**) and 4 (**Q4**) of 1- and 2-operator milkings across all parlor systems (Table 4). For 1-operator milkings, Q4 operators among RO parlors achieved cows/h values that were, on average, 95% greater than values achieved by Q1 operators. Lower average efficiency gaps were observed in H-DU parlors (+67%), H-SO parlors (+22%), and P-DU parlors (+18%). Similarly, for 2-operator milkings, Q4 operators among RO parlors achieved cows/h values that were, on average, 196% greater than values achieved by Q1 operators. Lower average efficiency gaps were observed in H-SO (+166%), H-DU (+76%), and P-DU parlors (+28%; Table 4). This variance in efficiency would be driven by factors related to parlor infrastructure, automation use, or management practices at milking. For example, Prendergast et al. (2024b) observed that the most efficient operators in H-SO parlors achieved cows/h per operator values that were 82% greater than values achieved by the least efficient operators. The authors reasoned that this higher efficiency was facilitated by these operators using larger sized, highly automated parlors and performing fewer work routines at milking when compared with operators who achieved average milking efficiency values. Specifically, Armstrong and Quick (1986) documented a greater milking efficiency (+9% cows/h) among herringbone parlors that used automatic cluster removers compared with those that did not. Among 2-operator milkings, Armstrong and Quick (1986) also observed that a circular work routine (i.e., 2 operators working clockwise/anticlockwise around a milking parlor, each carrying out different work elements) resulted in a 14% increase in cow throughput than an end-to-end work routine (i.e., 2 operators operating the same procedures but at opposing ends of the parlor).

Similarly, among RO parlors, Armstrong et al. (2001) reported that operators who carried out no premilking work routines achieved an average efficiency of 137 cows/h per operator. This value was 20% and 74% greater, respectively, than operators that carried out minimal (strip, wipe, attach) and full routines (predip, strip, wipe, attach). Andersson (2014) observed a 41% difference in milking efficiency (cows/h) between RO parlors with parallel angled clusters and 15-degree angled clusters. From a sample of New Zealand dairy farms using RO parlors, Edwards et al. (2020) reported that the use of an automatic teat sprayer at milking was associated with a 20% increase in cows/h per operator values.

The substantial variances in milking efficiency observed across different parlor types, for 1- and 2-operator milkings, highlights the influence of both parlor configuration and management practices on milking parlor performance. Existing research shows that the use of automation and standardized work routines can significantly increase cow throughput per hour. The findings highlighted in this review reinforce the perspective that optimum milking efficiency arises from the correct alignment of milking management practices, automation specification, and parlor infrastructure. Although considerable progress has been made in outlining strategies toward H-SO and RO parlor milking efficiency (Prendergast et al. 2025a,b), further research is required. Particularly, exploration is required into multiple-operator milking scenarios across different milking parlor types with alternative levels of infrastructure, automation, and management variables being accounted for. This knowledge will help to enhance the current understanding of milking parlor efficiency in the dairy industry across either PB or CB production systems.

Evolution of Research Design and Future Study Directions

It was found that "comparative" objectives were the most frequent problem type across the reviewed milking efficiency literature, comprising 53% of all examined publications. This was followed by "descriptive" and "technical development" problem types, which accounted for 27% and 20% of examined publications, respectively. Consequently, the current understanding of milking process efficiency is primarily based on studies that examined milking efficiency through comparison of different parlor types. Publications with descriptive problem types were detail-oriented and focused on quantifying effects of low-level parameters at milking, such as operator work routines, work routine times, operator working patterns, walking distances, walking times, automation variations, and cow milking times, on milking process efficiency. Publications with technical development problem types focused on the development and demonstration of a method or simulation with intent to identify strategies toward milking efficiency optimization. These publication types represented a minority in the literature. Although evidence exists of sufficient information in the literature for describing parlor performance, operator working patterns, and work routines, the current body of knowledge lacks empirically developed mechanisms that can allow for multiple milking efficiency scenario explorations. Therefore, future research would benefit from simulation-enhanced investigation where models are capable of evaluating multiple milking efficiency scenarios with respect to varying levels of parlor infrastructure, automation specification, and management practices at milking.

Limitations of the Study

This study was not without limitations. Primarily, the small number of included studies (n = 15) limits the generalizability of the modeled findings and likely reduces statistical power for pairwise comparisons. Therefore, the absence of significant differences for several KPI should not be interpreted as evidence of equivalence between parlor types but rather may reflect limited power to detect true effects.

In addition, the literature search was restricted to 2 primary databases. Although supplemented by reference list screening and targeted author searches, this approach may have increased the risk of selection bias and omission of relevant studies indexed elsewhere.

Furthermore, the findings of this study were built on the assumption that reported KPI were derived using a process time defined as first cluster attached until last cluster detached; however, for many studies this was not explicitly outlined by the authors. Consequently, differences in time-window definitions may affect KPI comparability. In addition, setup and cleanup activities, which can account for a meaningful proportion of labor input, were not systematically reported and therefore could not be incorporated into the analysis.

Moreover, limited reporting of automation specifications, operator routines, and management practices within the reviewed literature restricted the ability to stratify results by these potentially influential factors. As a result, residual heterogeneity across studies remains, limiting causal inference, reducing the strength of cross-type comparisons, and constraining the ability to quantify specific strategies for parlor efficiency optimization. In addition, individual KPI such as cows/h and clusters/operator are structurally influenced by parlor capacity and automation level, which may further affect their interpretation across systems.

Accordingly, this review should be interpreted primarily as a structured synthesis and comparative description of the existing literature, rather than a definitive prescriptive framework for parlor design or management.

Addressing the Research Questions

In relation to RQ1, the country and production system responsible for the largest contribution to milking efficiency literature was the United States and CB production systems. Regarding RQ2, the most common research problem identified among the literature was comparative, representing 53% of all examined publications. For RQ3, the most common methodology employed among milking efficiency research was time-and-motion analysis, which represented 60% of all identified methodologies. In relation to RQ4, P-DU parlors achieved the highest value for milking efficiency (352 cows/h), and RO parlors achieved the highest values for operator efficiency (143 cows/h per operator) and number of clusters handled by an operator at milking (27 clusters/operator). The H-SO parlors achieved the highest value for cows milked per cluster per hour (5.6 cows/h per cluster). For RQ5, we found that across all parlor types, increasing operator number did not significantly increase cow throughput ($P = 0.51$) or cows/h per cluster ($P = 0.64$) but significantly reduced cows/h per operator ($P < 0.005$) and clusters/operator ($P < 0.0005$) at milking.

CONCLUSIONS

We analyzed empirical evidence on milking parlor efficiency across 4 parlor types operating in both PB and CB production systems. Rotary parlors demonstrated the highest values for cows/h per operator, P-DU parlors yielded the greatest values for cows/h, and H-SO parlors achieved the highest values for cows/h per cluster. We identified the United States and CB systems as dominant contributors to the literature, with comparative studies and time-and-motion analyses forming the primary research approach. The evidence synthesized in this review suggests that milking efficiency is strongly influenced by the alignment of parlor infrastructure, automation specification, and management practices. However, heterogeneity in study design and KPI definitions limits causal interpretation. Future research would benefit from simulation-enhanced modeling approaches capable of evaluating multiple milking efficiency scenarios under varying infrastructure, automation, and management configurations.

NOTES

This study received no external funding. No human or animal subjects were used, so this analysis did not require approval by an Institutional Animal Care and Use Committee or Institutional Review Board. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AUS = Australia; CB = confinement based; H-DU = herringbone double-up; H-SO = herringbone swing-over; IRE = Ireland; IS = infrastructure survey; LAT = Latvia; MEX = Mexico; MMD = milking management data; MNGT-S = management survey; MYD = milk yield database; NCE = North Continental Europe; NZ = New Zealand; PB = pasture based; P-DU = parallel double-up; Q1 = quartile 1; Q4 = quartile 4; RO = rotary, RQ = research question; T&M = time-and-motion analysis; TEL-S = telephone survey; UK = United Kingdom; US = United States; VR = video-recording.

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